

Knowledge-anchored Sparse Autoencoders for Interpreting MedVLMs

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Introduction

Medical Vision-Language Models (Medical VLMs)

Definition:
Given a medVLM that has /alerthuman annotated feature representation, designing an interpretability method that that illustrate concept-related features and preserves output quality as well

Example: Given a frontal chest X-ray and the prompt *"Describe the findings in this image"*, a model such as LLaVA-Med or CheXagent generates free-form text such as: *"There is increased opacity in the right lower lobe consistent with consolidation. No pleural effusion or pneumothorax."*

Goal of interpretability for Medical VLMs:
Not to improve accuracy per se, but to make the model's internal computations inspectable — so that when the model generates a finding, we can identify *which internal features* caused it, and whether the same features cause hallucinated findings when they should not.

Objective

Medical Vision-Language Models (Medical VLMs)

- Performance with good accuracy
- Pixel attribution map which is human understandable
- Textural feature list also human understandable

Problem: SAEs trained in the standard unsupervised way (Bricken et al. 2023; Pach et al. 2025) produce features whose meanings must be *guessed* post-hoc by inspecting top-activating images — a procedure known to be unreliable (Stevens et al., 2025).

Task: Determine whether biasing SAE training with the symbolic concept set $\mathcal{C}$ as an auxiliary signal yields features that are (a) more reliably aligned with named clinical concepts, (b) causally faithful under intervention, and (c) useful for explaining specific hallucination types in medical VLM outputs.

Motivation

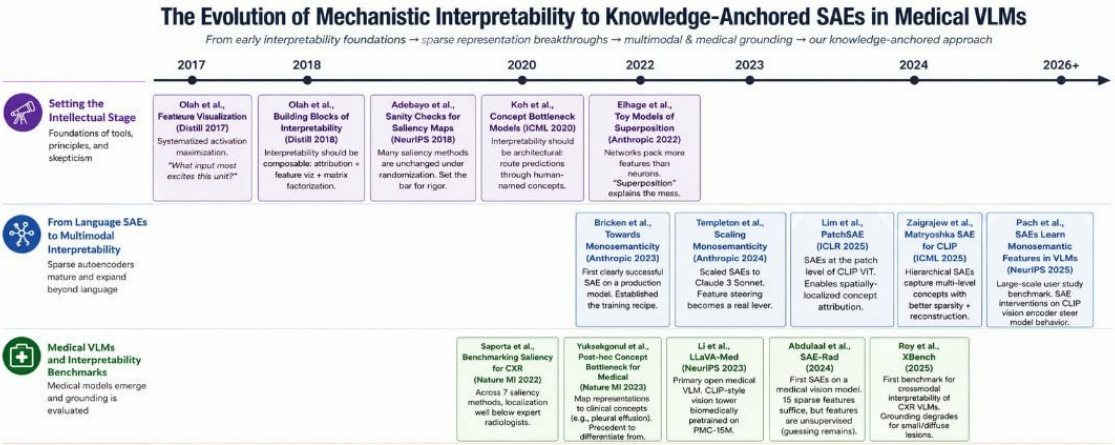
Our inspiration to persue the topic

- Images (X-rays, CT, MRI) + text (reports, notes) always come together
- Traditional models handle one modality at a time
  - VLMs jointly understand image and language, which matches real clinical workflow

Why interpretability in Medical VLMs is critical?

- Patient safety (non-negotiable)
  - MedVLMs can hallucinate findings
  - Doctors are clinically responsible for the patient, but the models are not.
  - Increasing amount of dependency on technologies in the medical field

Literature Review — Taxonomy and Chronology



Dataset: MIMIC-CXR

Overview:

- ∼377K chest X-ray images from ∼227K studies
- ∼65K patients (de-identified ICU population)
- Each study paired with free-text radiology report
- Typically 1–2 images per study (frontal/lateral views)

Why MIMIC-CXR?

- Supports multimodal learning (Image & Text)
- CheXpert 14-label schema for standardized classification
- Unstructured reports enable clinical concept extraction (e.g., Rad-Graph)
- Enables alignment between visual features and medical concepts

Dual-encoder/ Contrasive Strategy

Key Takeaways

- By decoupling images and texts, MedCLIP can scale usable training data **combinatorially**, allowing a small number of images and reports to generate a massive amount of supervision signal.
- The model uses UMLS entities and medical knowledge to identify when reports from different patients describe the same findings, preventing the model from being confused by **noisy** negative samples.

Summary

Best data efficiency when paired data is scarce; strongest at zero-shot classification without needing dense labels.

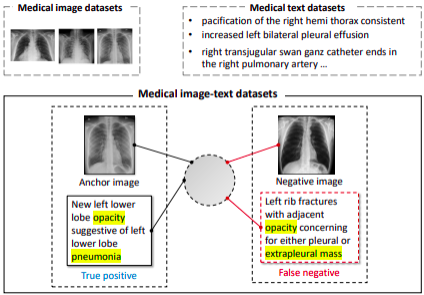
Advantages

- Can utilize unpaired medical datasets (image-only or text-only), which are more abundant than paired data.

Disadvantages

- Occasionally fails to correctly detect negation or uncertainty phrases within raw reports.

Illustration



M3AE: Multi-Modal Masked Autoencoders (Olah et al., 2017)

Key Takeaways

- It uses a high masking ratio for images (75%) and a lower ratio for text (15%) to account for the spatial redundancy in visual data
- The model captures hierarchical information by selecting intermediate-layer features for image reconstruction while using final-layer features for language
- It achieved state-of-the-art results on a new benchmark covering VQA, image-text classification, and retrieval tasks

Summary

M3AE is best at learning general-purpose multimodal representations without annotation; good pretraining backbone.

Advantages

- Does not require expensive manual annotations for pre-training; effectively models the cross-modal mapping between medical images and reports.

Disadvantages

- Performance is sensitive to the specific layer chosen for reconstruction; masking image pixels alone (without language masking) does not improve performance.

Illustration

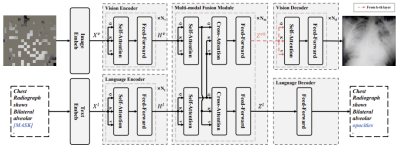


Fig. 1. The overall architecture of our proposed approach, where the vision encoder, language encoder, multi-modal fusion module, and decoders are shown in dash boxes.

LLaVA-Med: Large Language-and-Vision Assistant for BioMedicine

Key Takeaways

- It demonstrates an emergent multilingual capability, correctly answering Chinese questions despite being trained primarily on English data.
- Best at open-ended interpretable dialogue across diverse medical modalities; most flexible for follow-up tasks like our interpretability study.

Summary

LLaVA-Med is a cost-efficient large multimodal model (LMM) designed to answer open-ended research questions about biomedical images. It was developed by fine-tuning a general-domain LLaVA model using a curriculum learning approach: first aligning biomedical vocabulary through large-scale figure-caption pairs, and then mastering conversational semantics using instruction-following data generated by GPT-4.

Advantages:

- High efficiency in training; excellent multimodal conversational capabilities; outperforms supervised state-of-the-art on standard biomedical VQA metrics.

Disadvantages:

Still prone to **hallucinations** and lacks deep reasoning compared to human experts; performance is limited on open-set questions that lack constrained answer options.

Illustration

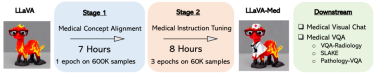


Figure 3: LLaVA-Med was initialized with the general-domain LLaVA and then continuously trained in a curriculum learning fashion (first biomedical concept alignment then full-blown instruction-tuning). We evaluated LLaVA-Med on standard visual conversation and question answering tasks.

XBench: Benchmark for Visual-Language Explanations

Key Takeaways

- Models trained on CXR-specific datasets demonstrate significantly better grounding than those trained on general-domain images.

Summary

XBench is the first systematic benchmark for evaluating cross-modal interpretability (grounding) in chest X-rays across various CLIP-style vision-language models. It assesses how well a model's visual **attention** aligns with regions annotated by human radiologists across 36 different findings.

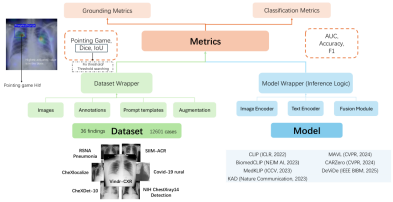
Advantages:

- Provides a rigorous foundation for validating model interpretability; identifies a critical need for score calibration in medical models.

Disadvantages:

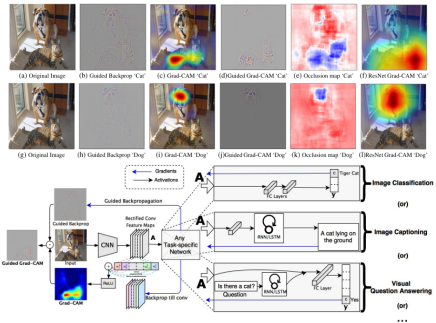
- Reveals that even the best current models fall short on small or diffuse lesions; models often rely too heavily on global context rather than local visual evidence.

Illustration



Grad-CAM (Visual Explanations via Gradient Localization)

Illustration



Takeaway

Grad-CAM enables class-specific visual explanations by highlighting image regions that positively influence model predictions, even in multimodal settings.

Summary

Grad-CAM computes neuron importance weights using gradients flowing into the final convolutional layer, forming localization maps via weighted activations and ReLU filtering. Extensions like Guided Grad-CAM provide high-resolution details, while the method generalizes to multimodal tasks such as captioning and VQA, enabling both visual and textual explanations.

Advantages

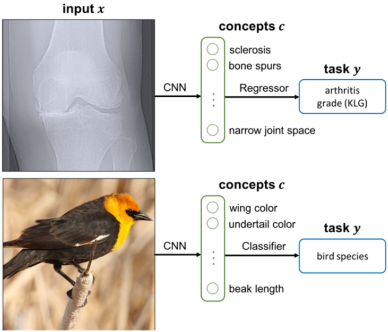
- Works without retraining or architecture changes.
- Applicable to diverse tasks, including multimodal models.
- Provides intuitive visual and textual explanations.
- Helps diagnose errors and uncover dataset bias.

Disadvantages

- Produces coarse, low-resolution heatmaps.
- Sensitive to layer selection.
- May miss negative or suppressive features.

Concept Bottleneck Models (CBMs)

Illustration



Takeaway

CBMs introduce a human-interpretable concept layer, enabling transparent reasoning and direct intervention without sacrificing model performance.

Summary

Concept Bottleneck Models decompose prediction into $x \to c \to y$, where c represents human-defined concepts. The framework supports multiple training strategies (independent, sequential, joint) and enables test-time interventions. CBMs improve robustness by relying on semantic features rather than spurious correlations.

Advantages

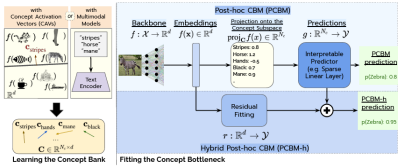
- Provides interpretable intermediate representations.
- Enables human-in-the-loop corrections.
- Improves robustness to spurious correlations.
- Achieves accuracy comparable to black-box models.

Disadvantages

- Requires predefined concept annotations.
- Limited by completeness of concept set.
- Concept prediction errors propagate downstream.

Post-hoc Concept Bottleneck Models (PCBM)

Illustration



Takeaway

PCBMs enable interpretable concept-based reasoning in any pretrained model by projecting features onto human-understandable concepts—without requiring costly annotations.

Summary

The method converts pretrained models into Concept Bottleneck Models by projecting embeddings onto a concept bank derived from CAVs or multimodal models like CLIP. Using language-defined concepts, PCBMs avoid manual labeling. Hybrid variants (PCBM-h) add residual predictors to preserve accuracy, while enabling global model editing through concept-level interventions.

Advantages

- No need for dense concept annotations.
- Leverages language (CLIP) for scalable concept definition.
- Enables global model editing and bias correction.
- Maintains high accuracy via hybrid residual modeling.

Disadvantages

- Concept quality depends on chosen concept bank.
- Residual component reduces full interpretability.
- Limited if concepts are incomplete or misaligned.

Feature Visualization (Olah et al., 2017)

Takeaway

Simple optimization-based methods can produce surprisingly clear visualizations of what a network is looking for, exposing a hierarchy from low-level edges to high-level object concepts.

Summary

The paper systematizes activation maximization by using gradients to modify an input so that a target neuron or channel becomes highly active. It also introduces practical improvements—such as regularization and diversity terms—to reduce adversarial artifacts and reveal multiple facets of the same learned feature.

Advantages

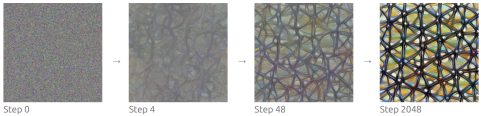
- Answers *what* the model is looking for.
- Reveals internal hierarchies from textures to objects.
- Helps surface spurious cues or bias before deployment.

Disadvantages

- Can produce high-frequency noise or "cheating" patterns.
- Results depend heavily on regularization choices.
- Visualizations may still be subjective or misleading.

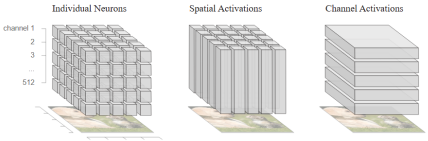
Illustration

Starting from random noise, we optimize an image to activate a particular neuron (layer mixed4a, unit 11).



Building Blocks of Interpretability (Distill, 2018)

Illustration



The cube of activations that a neural network for computer vision develops at each hidden layer. Different slices of the cube allow us to target the activations of individual neurons, spatial positions, or channels.

Takeaway

Interpretability can be decomposed into modular building blocks, combining visualization, attribution, and grouping to make neural representations more human-understandable.

Summary

The paper integrates multiple interpretability techniques into a unified interface. It introduces semantic dictionaries for mapping neuron groups to concepts, layer-wise attribution to trace feature contributions, and dimensionality reduction methods like NMF to compress neurons into interpretable groups.

Advantages

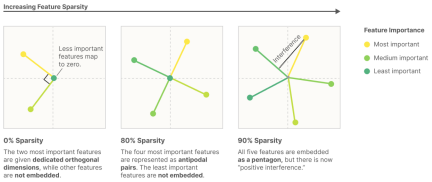
- Unifies multiple interpretability approaches.
- Keeps representations at a human-understandable scale.
- Highlights polysemantic and "alien" concepts.

Disadvantages

- Concept labels are lossy summaries.
- Grouping can obscure fine-grained mechanisms.
- Limited causal interpretability.

Toy Models of Superposition (Anthropic, 2022)

Illustration



Takeaway

Polysemantic neurons arise naturally when models compress many sparse features into limited neural capacity, leading to superposition.

Summary

The paper explains that neural networks with nonlinearities (e.g., ReLU) can encode more features than neurons by storing them in superposition. As sparsity increases, representations shift from one-feature-per-neuron to shared representations, forming structured geometric patterns like antipodal pairs and polytopes.

Advantages

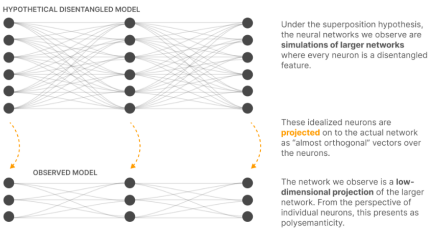
- Provides a theoretical basis for polysemanticity.
- Explains limits of neuron-level interpretability.
- Motivates feature-based analysis approaches.

Disadvantages

- Based on simplified toy models.
- Hard to directly apply to large real models.
- Geometry may not fully transfer to VLMs.

Towards Monosemanticity (Anthropic, 2023)

Illustration



Takeaway

Sparse autoencoders can partially disentangle superposed representations, moving neural features closer to monosemantic, interpretable units.

Summary

The work applies dictionary learning via sparse autoencoders (SAEs) to transformer activations, decomposing them into an overcomplete set of sparse features. Techniques like neuron resampling improve feature coverage, while increasing dictionary size leads to feature splitting, where broad concepts break into more refined subfeatures.

Advantages

- Reframes superposition as a solvable engineering problem.
- Produces features usable for causal intervention.
- Enables scalable and automated explanations.

Disadvantages

- Computationally expensive to train.
- Extracted features require careful validation.
- Interpretations can still be ambiguous.

SAEs Learn Monosemantic Features in VLMs (2025)

Illustration

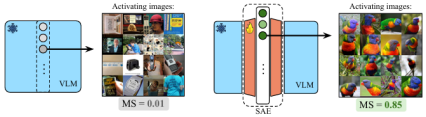


Figure 1: Sparse Autoencoder (SAE) in VLM (e.g. CLIP): Top activating images of a neuron in a pretrained VLM layer are polysemantic (left), and those of a neuron in a sparse latent of SAE trained to reconstruct the same layer are monosemantic (right), according to MonoSemanticity score (MS).



Figure 3: Top activating images of neurons with MonoSemanticity (MS) scores ranging from high (left) to low (right). Higher scores correlate with more similar images, reflecting monosemanticity.

Takeaway

Sparse Autoencoders (SAEs) can transform polysemantic VLM representations into interpretable, monosemantic features that are both measurable and causally controllable.

Summary

The paper extends SAEs to vision-language models (e.g., CLIP, SigLIP) to decompose activations into sparse, higher-dimensional features. It introduces the MonoSemanticity (MS) score to quantify interpretability and validates it with human studies. Advanced SAE variants (e.g., BatchTopK, Matryoshka) improve feature quality, while interventions on SAE features enable steering of downstream multimodal LLM outputs.

Advantages

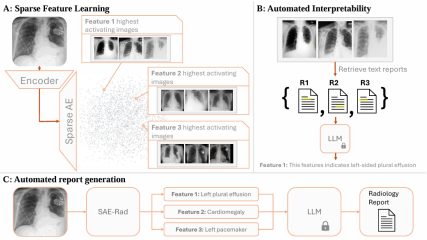
- Provides a quantitative interpretability metric (MS score).
- Enables unsupervised discovery of concept dictionaries.
- Supports causal steering of multimodal outputs.
- Validates superposition in VLM representations.

Disadvantages

- Computationally expensive (large SAEs, human studies).
- Interpretations still require validation.
- Scaling to very large VLMs remains challenging.

SAE-Rad (Sparse Autoencoders for Radiology)

Illustration



Takeaway

SAE-Rad transforms neural activations into interpretable clinical features, enabling transparent and efficient radiology report generation without heavy fine-tuning.

Summary

The framework uses a hybrid sparse autoencoder to extract sparse, clinically meaningful features from medical images. These features are automatically labeled using an LLM and combined to generate reports, avoiding traditional end-to-end training. Counterfactual interventions validate that the features are causally meaningful and faithful to the model.

Advantages

- Enables interpretable, feature-based report generation.
- Requires significantly less compute and parameters.
- Produces clinically faithful and low-error outputs.
- Supports causal validation via counterfactuals.

Disadvantages

- Not optimized for natural language fluency.
- Depends on quality of feature labeling by LLMs.
- Limited to domains with structured visual patterns.

RadGraph (Jain et al., 2021)

Key Takeaway

RadGraph converts free-text radiology reports into structured entities and relations, helping connect unstructured clinical text with automated medical image interpretation.

Summary

The paper introduces a radiology knowledge-graph schema with key entity types such as anatomy and observations, plus relations such as located at and modify. It also releases a large benchmark with millions of extracted entities and relations, making it useful for chest X-ray report understanding and multimodal learning.

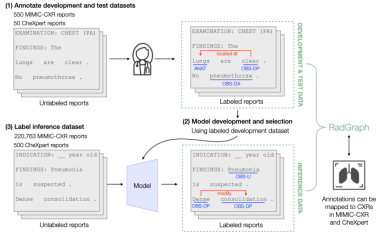
Advantages

- Captures rich clinical structure beyond disease labels.
- Supports fine-grained text–image learning for CXR.
- Large expert-annotated resource improves reliability.

Disadvantages

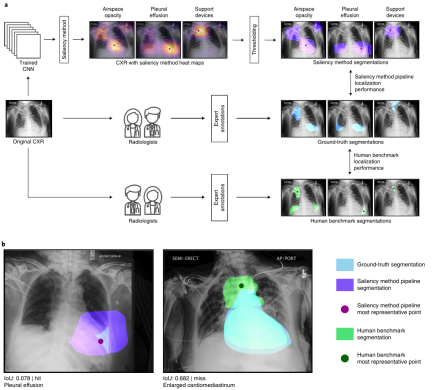
- Annotation is costly and requires radiology expertise.
- Focus is mainly on chest X-ray reports.
- Relation labels can still be ambiguous.

Illustration



Benchmarking Saliency Methods for CXR

Illustration



Takeaway

Common saliency methods often fail to reliably explain model decisions, performing significantly worse than human experts in high-stakes medical imaging.

Summary

The paper systematically evaluates seven saliency methods (e.g., Grad-CAM, Integrated Gradients, LRP) across multiple CNN architectures using a human-annotated benchmark (CheXlocalize). Localization is measured using strict (mIoU) and relaxed (Hit Rate) metrics, revealing a substantial gap between automated explanations and expert annotations.

Advantages

- Provides rigorous, quantitative evaluation of interpretability methods.
- Introduces expert-annotated benchmark (CheXlocalize).
- Identifies failure modes (small/irregular structures).
- Highlights confidence–interpretability correlation.

Disadvantages

- Limited to CNN-based medical imaging.
- Saliency methods remain coarse and low-resolution.
- Results depend on chosen evaluation metrics.

Conclusion

Challenges in SAE-Rad

- **Feature names are guessed, not verified.** An LLM infers each feature's meaning from top-activating images — the same failure mode Stevens et al. (2025) exposed, where a "Brazil" feature was actually firing on lamps.
- **No spatial grounding.** Uses only the class token, so findings cannot be localized within the image.
- **No causal check.** Feature identity is asserted, never tested by ablation or intervention.

How our approach addresses these

- **Names by supervision, not by guessing.** RadGraph labels anchor feature identity at training time — each anchored feature is optimized to fire on its named concept and nowhere else.
- **Patch-level architecture.** Features attribute to specific image regions, enabling spatial grounding evaluable against radiologist segmentations.
- **Causally verifiable.** Ablating a feature must change the VLM's output for that concept specifically — identity is tested, not assumed.

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